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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

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PHYSICS **5054/43**

Paper 4 Alternative to Practical **October/November 2010**

1 hour

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of **11** printed pages and **1** blank page.

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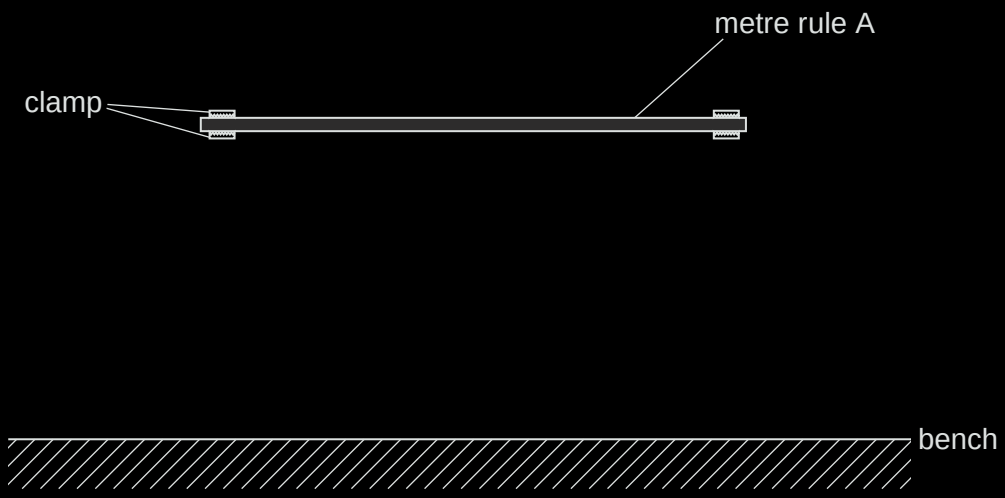


Fig. 1.1

(a) Describe how the student checks that A is horizontal. You may draw on Fig. 1.1.

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..... [1]

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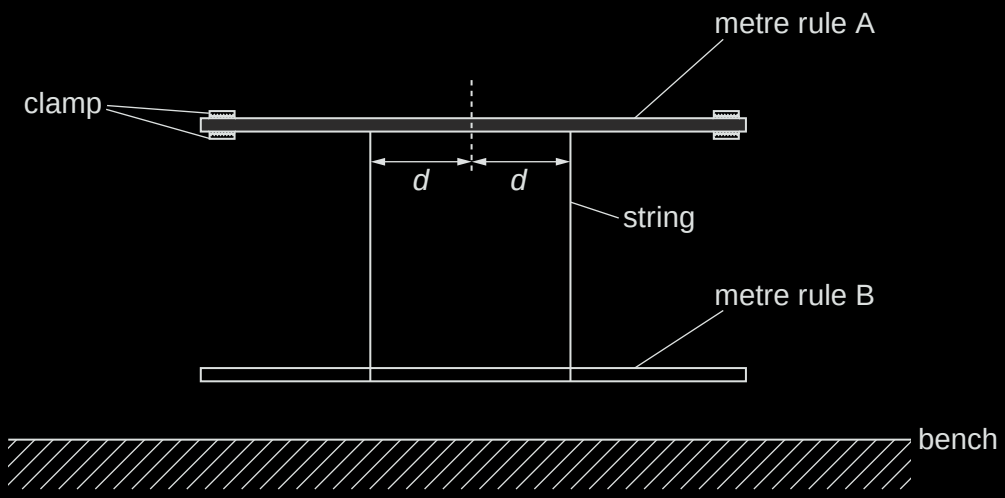


Fig. 1.2

Describe how the student checks that the two strings are equal distances from the centre of metre rule B.

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.....

.....[1]

(c) B is made to rotate backwards and forwards through a small angle in a horizontal plane about its centre, as shown in Fig. 1.3.

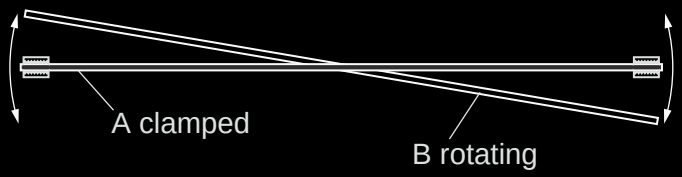


Fig. 1.3 (view from above)

The student determines the time *T* for B to rotate once, forwards and then backwards.

(i) **On Fig. 1.2**, mark where the eye of the student is placed to observe the rotations. [1]

(ii) Describe how the student determines *T* accurately.

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.....[3]

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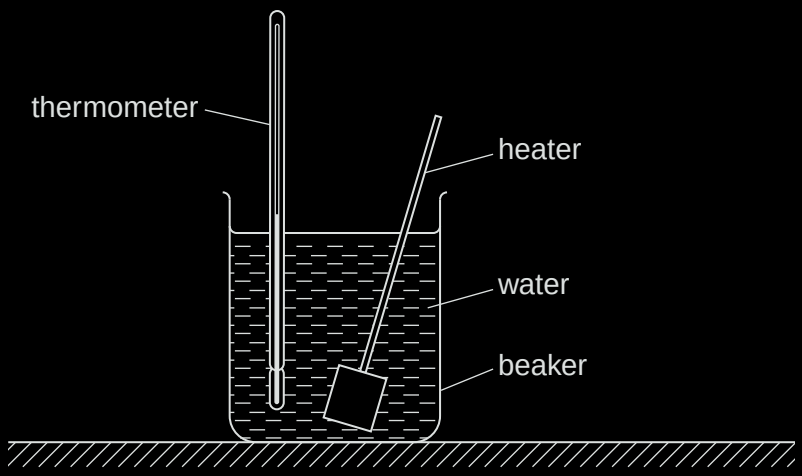


Fig. 2.1

The student measures the temperature of the water every minute for thirty minutes.

(a) The student suggests using the second hand of a wall clock to measure the time. Another student suggests using a stopwatch reading to 0.01 s. Suggest and explain which timer is the more suitable to use.

.....

..... [1]

(b) The student draws a table to record the results.

On Fig. 2.2, complete the headings of the table.

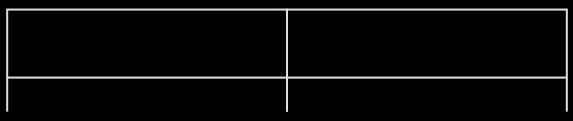


Fig. 2.2

[2]

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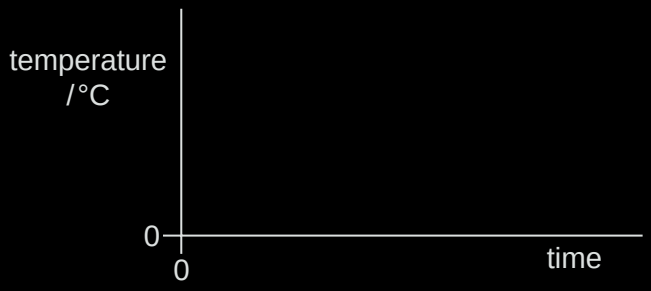


Fig. 2.3

[3]

(ii) Explain why the temperature of the water does not rise above 60°C.

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..... [1]

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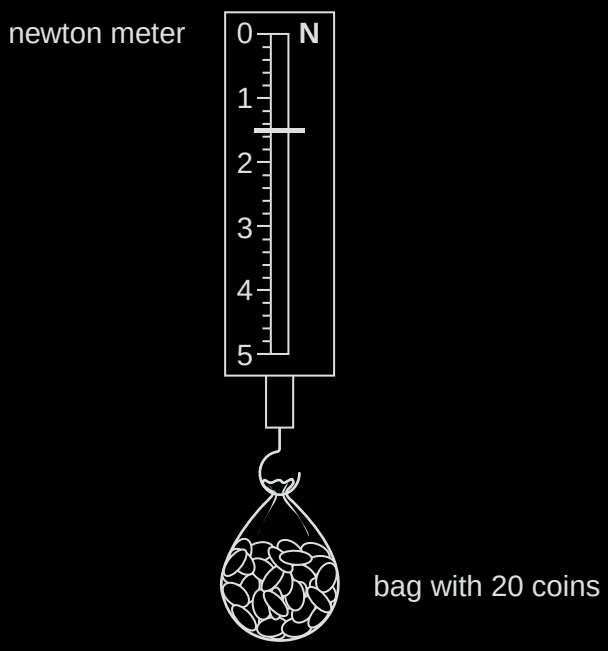


Fig. 3.1

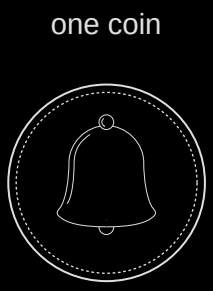


Fig. 3.2 (actual size)

(a) Record the reading from the newton meter to find the weight W_{20} of the 20 coins.

W_{20} = ... N [1]

(b) A full-size diagram of a single coin is shown in Fig. 3.2.

Measure accurately the diameter d of the coin.

d = ... cm [1]

(c) The average thickness t of a coin is 0.19 cm.

Describe how to measure t accurately.

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.....[2]

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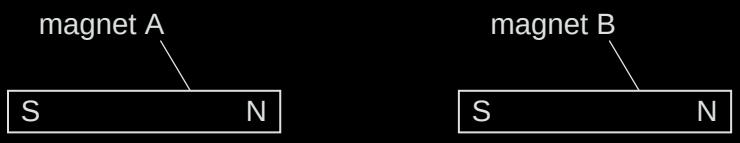


Fig. 4.1

The student is given a list of apparatus available for use in the experiment.

List of available apparatus

metre rule	magnet	pencil
20 cm ruler	newton meter	plain paper
clamp stand	plotting compass	lined paper
clamp	iron bar	graph paper
boss	iron paper clips	thin cotton

(a) Describe how to determine which magnet is the stronger.
You should include a diagram to show how your apparatus is used.

Apparatus chosen: .....

Diagram:

Explanation:

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..... [3]

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